CSF3133 • TOPIC 01

INTRODUCTION TO INTERNET AND WORLD WIDE WEB

COURSE LECTURER

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COURSE FRAMEWORK

Learning Objectives

- Describe the evolution of the Internet and the Web
- Explain the need for web standards & Universal Design
- Identify benefits of accessible web design & ethics
- Describe the purpose of web browsers and web servers
- Identify networking protocols, URIs, and domain names
- Describe HTML, XHTML, HTML5, and popular digital trends

OPENING REFLECTION

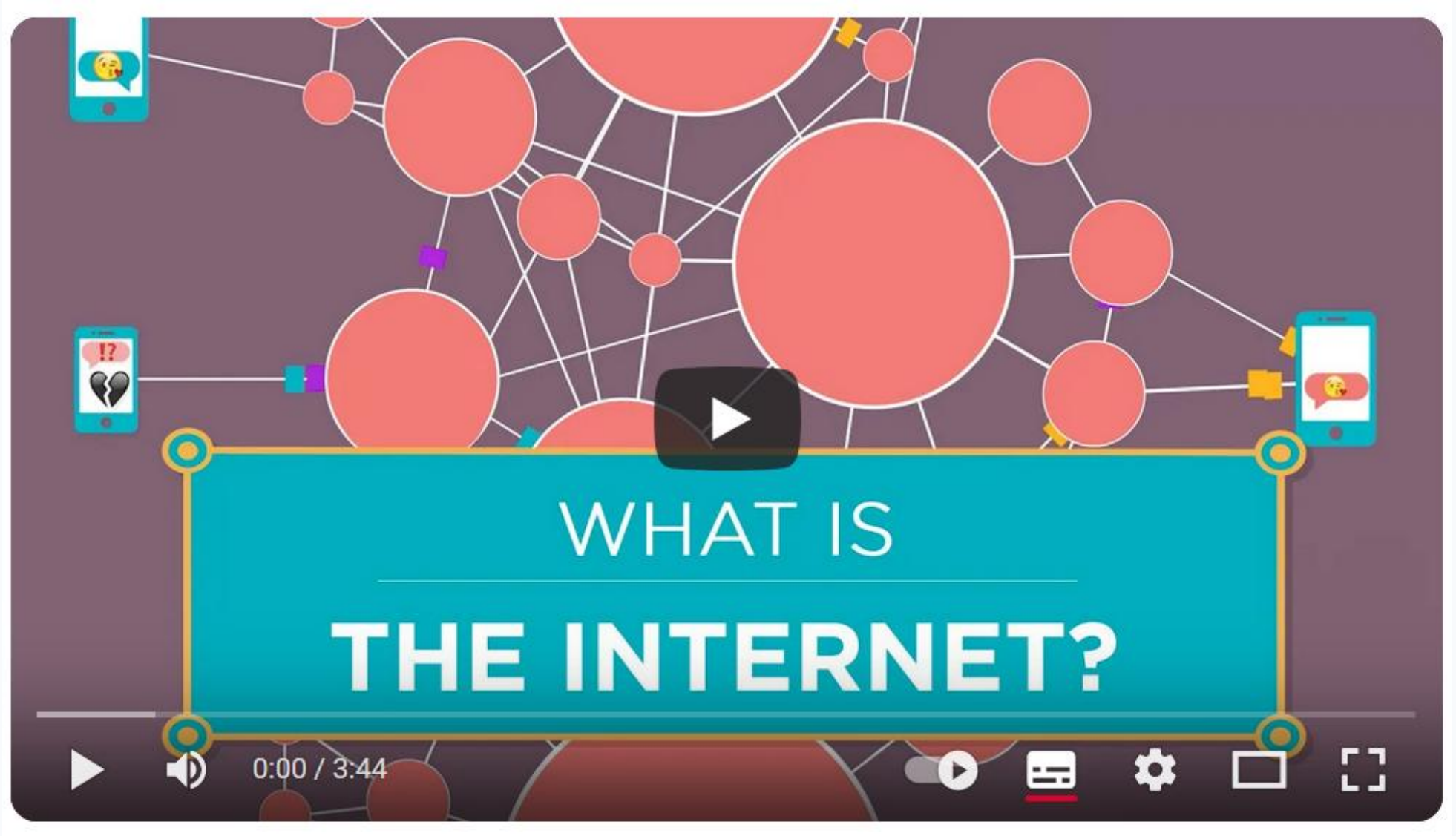
Reflective Thought

“Do you remember your first experience using the Internet?”

Reflect on how much the medium has changed your day-to-day interactions.

MULTIMEDIA RESOURCE

What is the Internet?



Reasons for Internet Growth in the 1990s

The 1990s saw rapid growth in Internet usage due to **commercial opportunities**, the **World Wide Web**, user-friendly browsers and **affordable personal computers.**

1 Commercial Activity

Removal of restrictions on **commercial traffic** by the US government.



- In the early years, the Internet was mainly for research and educational use.
- In 1991, the US government lifted restrictions on commercial traffic, allowing businesses to use the Internet.
- This opened the door for e-commerce, online services and many new business opportunities.

2 World Wide Web

Invention of HTML and HTTP by Tim Berners-Lee at CERN.

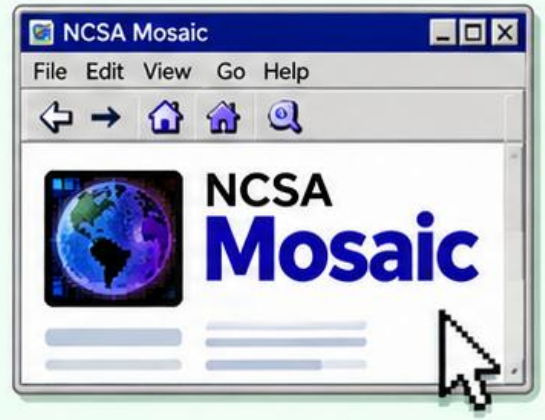


Tim Berners-Lee
CERN (1989–1991)

- Tim Berners-Lee developed the World Wide Web at CERN.
- He created **HTML** (Hypertext Markup Language) to structure web pages.
- He also developed **HTTP** (Hypertext Transfer Protocol) to enable communication between web browsers and web servers.
- The Web made the Internet easier to use and accessible to everyone.

3 Mosaic Browser

First graphical interface browser developed at NCSA.



- Mosaic, developed in 1993 at the National Center for Supercomputing Applications (NCSA), was the first widely popular graphical web browser.
- It allowed users to view text and images in a single, easy-to-use interface.
- Mosaic made the Web visually appealing and user-friendly, which attracted millions of new users to the Internet.

4 Affordable PCs

Widespread adoption of low-cost personal computers & dial-up ISPs.



- In the 1990s, personal computers became more affordable and widely available to homes and small businesses.
- Internet Service Providers (ISPs) offered dial-up connections, making Internet access easier and cheaper.
- This led to a rapid increase in the number of Internet users around the world.

Combined Impact

These four factors worked together to make the Internet **more accessible, useful** and **attractive,** leading to **rapid global growth** in the 1990s.

Key Layers of the Internet

Early Milestones

- **1971** email @ Ray Tomlinson
- **1990** Archie (Emtage & Deutsch)
- **1986** DOS Houdini (Neil Larson)
- (Vannevar Bush, Ted Nelson, Douglas Engelbart)
- **1969** ARPANET (J.C.R. Licklider)
- **1956** SAGE (George Valley)
- **1941** Z3 (Konrad Zuse)



Later Milestones

- **1987** HyperCard (Bill Atkinson)
- **1998** Google (Brin & Page)
- **1993** Mosaic (Marc Andreessen)
- **1990** http:// (Tim Berners-Lee)
- **1975** TCP/IP (Cerf & Kahn)
- **1973** Ethernet (Robert Metcalfe)
- **1976** Apple (Jobs & Wozniak)

ORIGINS

ARPANET (1970s Infrastructure)

CORE CONCEPT

Packet-Switching Pioneer

In the 1970s, ARPA's network connected research universities and defense nodes across the United States, laying the architectural groundwork for today's Internet.

GLOBAL STATISTICS

Essential Digital Headlines (2025)

ITEM 1

8.20 Billion

Total World Population (58.1% Urbanisation)

ITEM 2

5.78 Billion

Unique Mobile Phone Subscribers (70.5% of population)

ITEM 3

5.56 Billion

Active Internet Users (67.9% of global population)

ITEM 4

5.24 Billion

Social Media User Identities (63.9% adoption)

Main Reasons for Using the Internet

People use the Internet for many purposes. Here are the main reasons.

The Internet helps us **communicate, find information, be entertained, and learn or work more effectively.**

1 Communication

Staying in touch with family, colleagues, and social networks.



Examples of online communication:

- Email (e.g. Gmail, Outlook)
- Messaging apps (e.g. WhatsApp, Telegram)
- Video calls (e.g. Zoom, Microsoft Teams)
- Social media (e.g. Facebook, Instagram)

2 Information Search

Looking up news, academic topics, recipes, and references.



Examples of information search:

- Latest news and current events
- Academic topics and research
- Recipes, health and lifestyle information
- References, tutorials and how-to guides

3 Entertainment

Streaming video, music, interactive media, and games.



Examples of online entertainment:

- Streaming video (e.g. YouTube, Netflix)
- Music streaming (e.g. Spotify, Apple Music)
- Online games (e.g. mobile and PC games)
- Podcasts, digital media and live streaming

4 Education & Work

Online learning, remote workflows, productivity, and e-commerce.



Examples of education and work:

- Online learning (e.g. MOOCs, LMS)
- Remote work and collaboration (e.g. Google Workspace, Microsoft 365)
- Productivity tools (e.g. Docs, Sheets, Trello)
- E-commerce and online services (e.g. online shopping, banking)

Key Takeaway

The Internet has become an essential part of daily life because it helps us **communicate, access information, be entertained,** and **learn or work more effectively.**

Internet vs. World Wide Web

The Internet is the global network, while the World Wide Web is a service on the Internet.

The Internet

The global network of networks

The Internet is a worldwide system of interconnected computer networks that communicate with each other using standard protocols (e.g. TCP/IP).



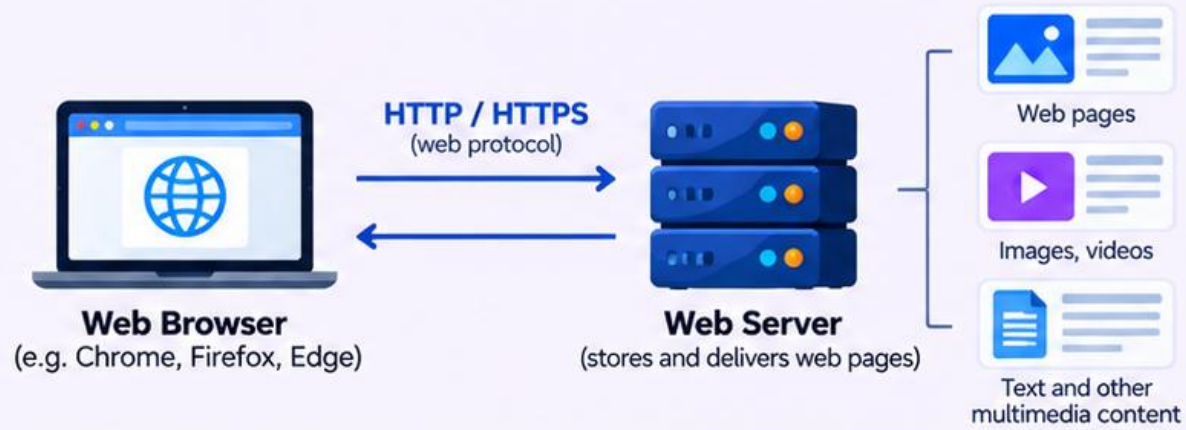
Key Points about the Internet

1. A global network of networks.
2. Connects millions of devices worldwide.
3. Uses communication protocols (e.g. TCP/IP).
4. Provides many services (web, email, file transfer, video streaming, etc.).

The World Wide Web

A service on the Internet

The World Wide Web (WWW) is an information system of interlinked web pages and resources that can be accessed using a web browser. It runs on top of the Internet.



Key Points about the World Wide Web

1. A collection of interlinked web pages and resources.
2. Accessed using a web browser (e.g. Chrome, Firefox, Edge).
3. Uses HTTP/HTTPS to transfer web pages.
4. It is just one of the many services available on the Internet (others include email, file transfer, online games, etc.).

Key Takeaway

The **Internet** is the **infrastructure** (the global network), while the **World Wide Web** is a **service** that uses the Internet.

Growth of the Internet (until 2026)

From research networks to a global digital society, the Internet has grown rapidly, transforming how we communicate, work, learn and live.



1960s

Early Networks

The foundation of the Internet.



- 1969: ARPANET first connected four computers in the U.S.
- Developed for research and military use.

1970s

Protocols & Email

Key technologies enable networking.



- 1971: First email sent.
- 1973: TCP/IP concept introduced.
- Laying the groundwork for internetworking.

1980s

From Research to Public Access

Networks expand beyond research institutions.



- 1983: TCP/IP adopted as the standard protocol.
- Domain Name System (DNS) introduced.
- More organisations and universities join the network.

1990s

The World Wide Web

The Internet becomes widely available to the public.



- 1991: World Wide Web (WWW) launched.
- 1993: Mosaic web browser.
- Commercial Internet Service Providers (ISPs).
- Websites, search engines and online services emerge.

2000s

Web 2.0 & Social Media

A more interactive and user-generated Internet.



- Rise of social media platforms.
- User-generated content (blogs, wikis, social networks).
- E-commerce, online services and cloud computing.

2010s

Mobile & Cloud Computing

The Internet goes mobile and becomes part of daily life.



- Smartphones and mobile Internet usage surge.
- Cloud computing and app ecosystems grow.
- Streaming, online collaboration and digital services become mainstream.

2020s

A Connected World

The Internet becomes essential for work, learning and society.



- Massive growth in online services.
- Accelerated digital transformation due to the COVID-19 pandemic.
- More connected devices (IoT), 5G networks and AI applications.

2026

Towards a Smarter and More Inclusive Internet

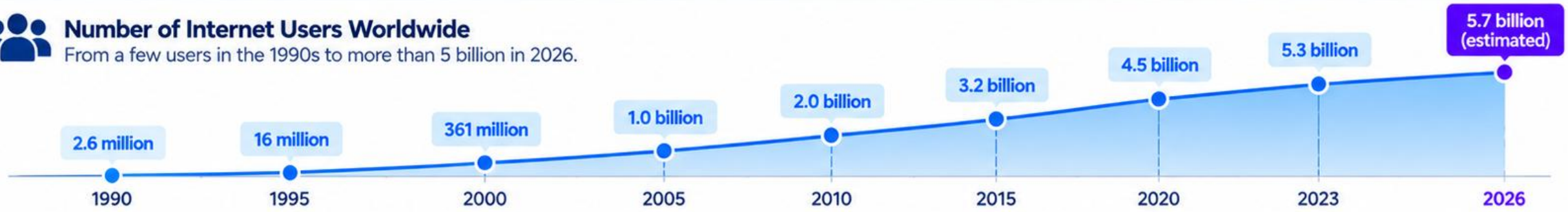
Continued growth with emerging technologies.



- Greater adoption of AI, IoT and 5G/6G.
- More personalised, immersive and intelligent services.
- Focus on digital inclusion, cybersecurity and sustainable Internet growth.

Number of Internet Users Worldwide

From a few users in the 1990s to more than 5 billion in 2026.



Source: ITU, World Bank, Statista (2023–2024), various sources. 2026 value is an estimate.

Internet Standards & Coordinating Bodies

Key organisations that develop and coordinate Internet standards and resources.



Open
Collaborative
Global
Interoperable

1 IETF

Internet Engineering Task Force

Develops **voluntary RFC network standards.**

IETF®

- Develops Internet standards in the form of RFCs (Request for Comments).
- Open, global community of engineers, researchers and operators.
- Standards are voluntary and developed through consensus.

Examples of work

TCP, IP, HTTP, SMTP, DNS and many other Internet protocols.

2 IRTF

Internet Research Task Force

Focuses on **long-term Internet research.**

IRTF®

- Explores emerging technologies and long-term research topics related to the Internet.
- Encourages experimental work and innovation.
- Organised into Research Groups (RGs) that study specific topics.

Examples of work

Future Internet architecture, Internet measurement, security, privacy and new networking paradigms.

3 IAB

Internet Architecture Board

Oversees **RFC publication and architecture.**

IAB

- Oversees the overall technical direction and architecture of the Internet.
- Manages the RFC Series and works with the IETF and IRTF.
- Advises on architectural issues and ensures long-term coherence of Internet standards.

Examples of work

Architecture oversight, RFC publication process, coordination between IETF and IRTF.

4 ICANN

Internet Corporation for Assigned Names and Numbers

Coordinates assignment of **domain names, IP addresses, and protocol port numbers.**

ICANN

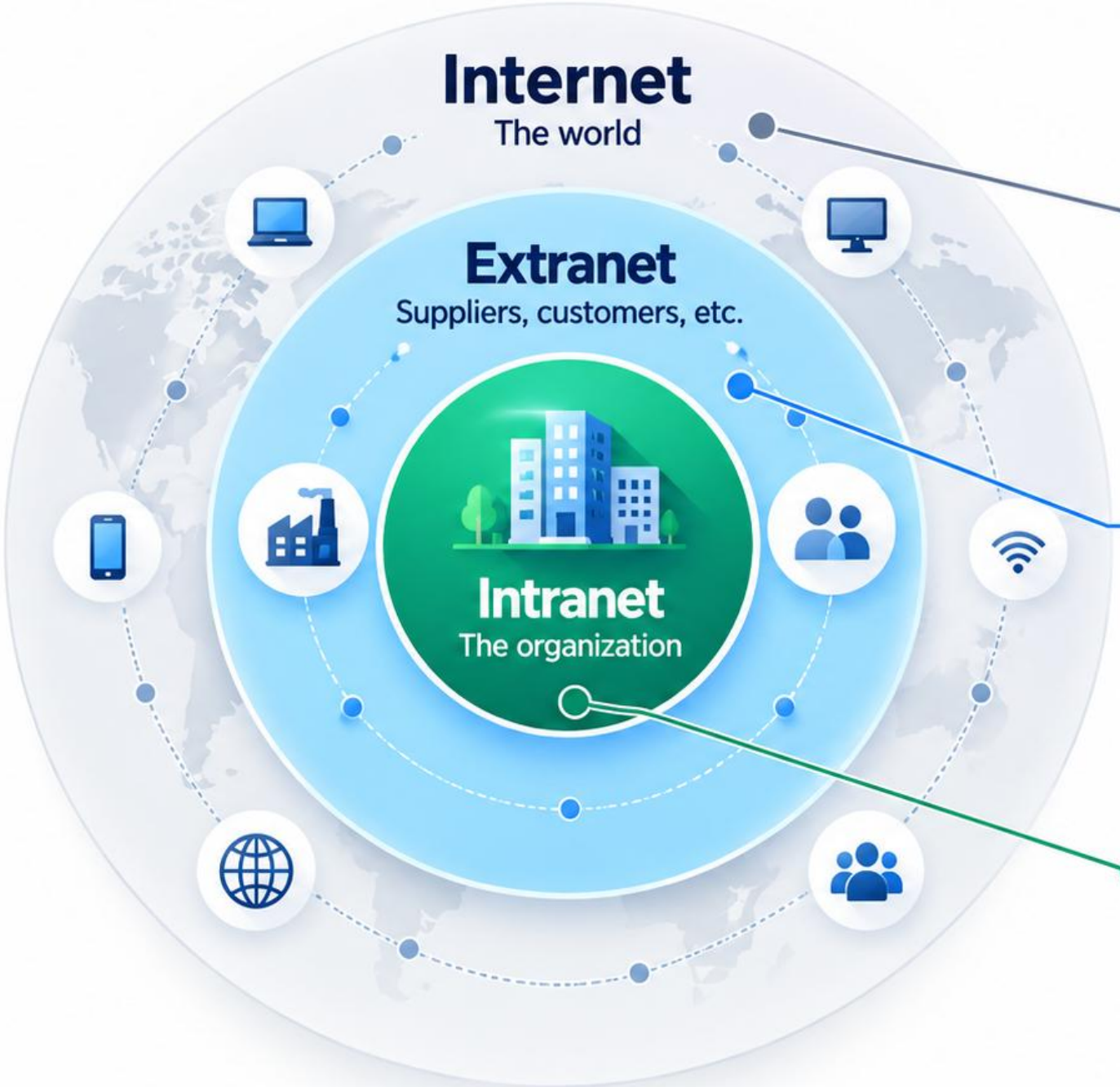
- Coordinates the global Internet's unique identifiers.
- Manages domain names (e.g. .com, .edu, .my), IP address allocation and protocol port numbers.
- Works with a global multistakeholder community, including governments, industry, technical experts and users.

Examples of work

Domain name system (DNS), IP address allocation (with RIRs), protocol port number management.

Key Takeaway

The IETF, IRTF, IAB and ICANN work together to develop, research, oversee and coordinate Internet standards and resources, helping to keep the Internet **open, interoperable and reliable.**



Internet

The **internet** creates connections between computers around the world.

Extranet

An **extranet** creates connections beyond (or outside) an organization to suppliers, customers, etc.

Intranet

An **intranet** creates connections inside an organization.

CORE PHILOSOPHY

Reflective Thought

“If we design for accessibility, everyone benefits.”

Inclusivity in digital architecture ensures seamless access regardless of physical ability or device constraint.

WEB STANDARDS

Web standards are a set of open specifications and guidelines for building web pages and web applications.

1 What are Web Standards?

Web standards are **open, internationally agreed specifications** that define how web content is created, structured, presented and behaves on the web.

2 Why are Web Standards Important?

- Ensure websites work consistently across different browsers and devices.
- Improve accessibility for all users.
- Promote interoperability and innovation.
- Make the web more reliable, secure and future-proof.

3 Examples of Web Standards

HTML
Defines the structure and content of web pages.

CSS
Defines the presentation and layout of web pages.

JavaScript
Defines the behaviour and interactivity of web pages.

4 Benefits of Using Web Standards

- Cross-platform compatibility
- Better accessibility
- More secure websites
- Encourage innovation

5 Web Standards in Action



By following web standards, the same website can be viewed consistently on different devices and browsers.

World Wide Web Consortium (W3C)

The international organisation that develops web standards.

1 What is the W3C?

The World Wide Web Consortium (W3C) is an international community that develops **open standards** to ensure the long-term growth of the Web.

2 Mission

To lead the Web to its full potential by developing standards, guidelines and tools that ensure a more open, accessible, secure and universally usable Web.

3 Key Activities

- Develop web standards (e.g. HTML, CSS, Web APIs, accessibility).
- Bring together industry, academia, governments and users.
- Provide guidelines, best practices and technical reports.
- Promote an open and inclusive Web for everyone.

4 How It Works

The W3C works through working groups made up of members from different organisations and communities.



5 Examples of W3C Standards

HTML · CSS · JavaScript · ARIA (accessibility) · SVG · Web API

Key Takeaway

Web standards, developed by the W3C, help make the Web **open, accessible, interoperable, secure** and work **consistently for everyone, everywhere**.

Web Accessibility Initiative (WAI)

Making the Web accessible to everyone.

The Web Accessibility Initiative (WAI) is a programme of the **World Wide Web Consortium (W3C)** that works to make the Web accessible for people with disabilities.

1 What is WAI?

The Web Accessibility Initiative (WAI) is a programme of the **W3C** that develops standards, guidelines, tools and resources to help **make the Web accessible** to people with disabilities.



2 Goals and Mission

WAI works to ensure that people with disabilities can **perceive, understand, navigate and interact** with web content, and that they can **contribute** to the Web.

Perceivable
Information can be perceived in different ways.

Understandable
Content and operation are easy to understand.

Navigable
Users can navigate and find content easily.

Robust
Content works reliably with a wide range of user agents and assistive technologies.

3 Key Outputs from WAI

WAI develops internationally recognised standards, guidelines and supporting resources, including:

WCAG	Web Content Accessibility Guidelines (explain how to make web content accessible)
ARIA	Accessible Rich Internet Applications (specifications for accessible web applications)
Authoring Tools	Guidance for developers and content authors
Educational Resources	Tutorials, techniques and examples
Research and Community Work	Collaboration with industry, academia, governments and people with disabilities

4 Web Content Accessibility Guidelines (WCAG)

WCAG provides guidelines for making web content more accessible. It is based on four key principles (POUR):

Perceivable
Information and user interface components must be presentable to users in ways they can perceive.

Operable
User interface components must be operable by all users.

Understandable
Information and the operation of the user interface must be understandable.

Robust
Content must be robust enough to be interpreted reliably by a wide variety of user agents, including assistive technologies.

5 How WAI Supports Different Users

Web accessibility helps many groups of users, including people with:

- **Mobility impairments** e.g. limited hand movement, use of assistive devices
- **Visual impairments** e.g. low vision, colour blindness, blindness (screen readers)
- **Hearing impairments** e.g. deaf or hard of hearing (captions, transcripts)
- **Cognitive impairments** e.g. learning disabilities, attention difficulties
- **Situational limitations** e.g. using a mobile device, noisy environment, low bandwidth

6 Benefits of Web Accessibility

Inclusive Web for Everyone
Allows more people to access and use your content.

Better User Experience
Clearer, easier to use and more user-friendly websites.

Wider Reach
Reaches a larger audience, including people with disabilities and older users.

Legal and Ethical Compliance
Supports accessibility laws and demonstrates a commitment to equal access.

Key Takeaway

The Web Accessibility Initiative (WAI) develops guidelines, standards and resources to help make the Web **accessible, inclusive** and **usable for everyone**.

Web Accessibility & The Law

Web accessibility is not just good practice — it is a legal and ethical responsibility. Laws and standards around the world require digital content to be accessible to everyone, including people with disabilities.

Accessible websites help organisations comply with the law, reduce the risk of discrimination, and ensure equal access to information and services for all.

1 Why It Matters

Equal Access
Everyone has the right to access information and services online.

Prevent Discrimination
Inaccessible websites can exclude people and may be considered discriminatory.

Legal Compliance
Many countries have laws and regulations that require web accessibility.

Wider Benefits
Accessible websites are easier to use for everyone and improve usability, SEO and brand reputation.

2 Key Laws and Regulations

Malaysia
- Persons with Disabilities Act 2008 (Act 685)
- Requires equal access to information and services, including digital platforms.
- Supported by the Malaysian Standard MS 2613:2019 (Web Accessibility).

United States
- Americans with Disabilities Act (ADA)
- Applies to public and private organisations.
- Websites and mobile apps must be accessible (e.g. WCAG compliance).

European Union
- European Accessibility Act (EAA) 2019/882
- Requires accessible digital products and services (including websites and apps).
- Based on EN 301 549 and WCAG 2.1.

United Kingdom
- Equality Act 2010
- Requires organisations to make reasonable adjustments to ensure digital accessibility.
- Public sector websites must meet WCAG 2.1 AA (under Public Sector Bodies (Websites and Mobile Applications) Regulations).

3 Standards and Guidelines

WCAG (Web Content Accessibility Guidelines)
- International standard developed by the W3C.
- Current version: WCAG 2.2.
- Three conformance levels: A, AA, AAA.

4 Consequences of Non-Compliance

Legal Action
Risk of complaints, lawsuits and fines.

Reputation Damage
Loss of public trust and negative brand perception.

Lost Opportunities
Excludes users and reduces potential customers or stakeholders.

Key Takeaway

Comply with relevant laws and follow accessibility standards (e.g. WCAG) to ensure a more **inclusive**, **fair** and **legally compliant** web.

Accessibility Verification Checks

Key checks to ensure your website is accessible to all users.

Use this checklist to verify that your web content follows accessibility best practices (e.g. WCAG).

1 Text Alternatives

Images and buttons provide descriptive alternative text.



Verification Checklist

- [ ] All informative images have meaningful alt text.
- [ ] Decorative images use alt =""
- [ ] Buttons with icons have descriptive labels or aria-label.

2 Color Contrast

Sufficient contrast ratio between text and background.



Verification Checklist

- [ ] Text has sufficient contrast (≥ 4.5:1 for normal text, ≥ 3:1 for large text).
- [ ] Information is not conveyed by color alone.
- [ ] Links are clearly distinguishable (e.g. underline or different color).

3 Keyboard Navigation

All links, forms, and buttons are operable without a mouse.



Verification Checklist

- [ ] All interactive elements can be reached using the Tab key.
- [ ] The focus indicator is visible (e.g. outline).
- [ ] Forms and menus can be completed using the keyboard (Enter, Space, arrow keys).

4 Clear Structure

Logical headings and clear focus indicators across pages.



Verification Checklist

- [ ] Headings are used in a logical and hierarchical order (h1, h2, h3).
- [ ] Page structure is clear and consistent across pages.
- [ ] A visible focus indicator shows which element is currently active.

Key Takeaway

These checks help make your website **perceivable**, **operable**, **understandable** and **robust** for all users, including people with disabilities.

Universal Design for the Web

Design web content that is usable and accessible by as many people as possible, regardless of their abilities, devices or situations.



A more inclusive web benefits **everyone**, creating a better experience for people of all abilities, devices and situations.

1 Equitable Use

Design that is useful and accessible to people with diverse abilities.



Example

- Provide text alternatives for images.
- Ensure all functions are available to all users.
- Avoid designs that only work for a specific group (e.g. mouse users).

2 Flexibility in Use

Accommodate a wide range of individual preferences and abilities.



Example

- Design responsive layouts for different devices and screen sizes.
- Support multiple input methods (keyboard, mouse, touch, voice).
- Allow users to adjust settings (e.g. text size, contrast).

3 Simple and Intuitive Use

Make the design easy to understand and use, regardless of experience, language or cognitive ability.



Example

- Use clear and consistent navigation.
- Use familiar icons and labels.
- Write in plain language.
- Provide clear instructions and feedback.

4 Perceptible Information

Present information in ways that can be perceived by all users, including those with sensory impairments.



Example

- Provide alt text for images.
- Include captions and transcripts for audio/ video content.
- Use sufficient color contrast.
- Do not rely on color alone to convey meaning.

5 Tolerance for Error

Minimize risks and help users recover from mistakes.

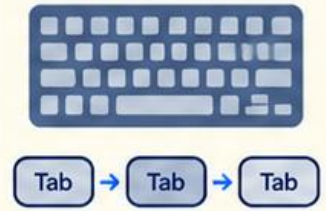


Example

- Provide helpful and clear error messages.
- Confirm important actions (e.g. before deleting data).
- Allow users to undo actions.
- Prevent and suggest solutions for common mistakes.

6 Low Physical Effort

Enable comfortable and efficient use with minimal physical effort.



Example

- Ensure all functions can be operated using a keyboard.
- Provide large, easy-to-click buttons and link areas.
- Avoid excessive time limits.
- Support voice input and other assistive technologies.

7 Size and Space for Approach

Provide comfortable size and space for interaction, regardless of device, body size or mobility limitations.



Example

- Use appropriately sized buttons, links and form fields.
- Provide enough spacing between interactive elements.
- Ensure content works well on different devices and orientations.
- Avoid tightly spaced clickable items.

Key Takeaway

Universal Design for the Web creates inclusive, user-friendly and accessible websites that work for **everyone**, in all situations.

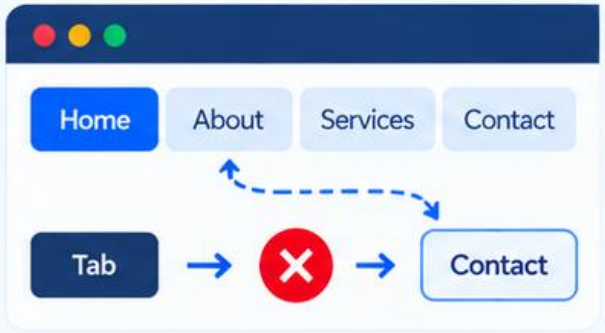
Common Digital Accessibility Issues

These issues can make it difficult or impossible for people with disabilities to access and use web content.

Identifying and fixing these issues helps create a more inclusive web for everyone.

1 Navigation Issues

Inconsistent or missing keyboard navigation paths.



Problem

Some menu items cannot be reached using the keyboard, or the focus order is inconsistent.

Better Approach

Ensure all links, menus, forms and buttons can be accessed in a logical order using the keyboard, with a clear focus indicator.

2 Color Contrast

Text unreadable against poor background contrast.

Read More — Low contrast (e.g. 1.8:1)

Text is difficult to read.

Read More — Good contrast (e.g. 7:1)

Text is clear and easy to read.

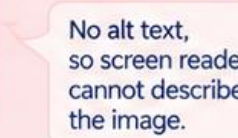
Better Approach

- Use sufficient color contrast (≥ 4.5:1 for normal text, ≥ 3:1 for large text).
- Avoid using color alone to convey important information.

3 Screen Readers

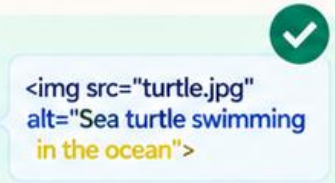
Missing semantic HTML tags and image alt attributes.





No alt text, so screen readers cannot describe the image.





Screen readers can describe the image to users.

Better Approach

- Use proper semantic HTML tags (e.g. <header>, <nav>, <main>, <footer>).
- Provide meaningful alt text for images and descriptive labels for buttons and links.

4 Media Barriers

Uncaptioned audio/video and auto-playing content.



No captions provided for video content.



Captions provided so that all users can access the content.

Better Approach

- Provide captions for all audio and video content.
- Avoid auto-playing media, or provide user controls to pause, stop or adjust the volume.

Key Takeaway

By addressing these common issues, we can create websites that are more **accessible, usable** and **inclusive** for all users, including people with disabilities.

Accessibility by Impairment Category

Different types of disabilities can affect how people access and use web content.

Designing with accessibility in mind helps create a **more inclusive web experience** for everyone.

Visual

Color blindness, low vision, and complete loss of sight.



Common Challenges

- Difficulty distinguishing colors.
- Small text is hard to read.
- Inability to see visual content (for screen reader users).

Design Considerations

- Use sufficient color contrast (≥ 4.5:1 for normal text).
- Provide text alternatives (alt text) for images.
- Use clear, readable fonts and allow text resizing.
- Do not rely on color alone to convey information.

Hearing

Deafness or hard-of-hearing requiring closed captions.



Common Challenges

- Cannot hear audio content.
- Important information may be missed in videos or audio.
- Rely on visual information (e.g. captions, transcripts).

Design Considerations

- Provide accurate closed captions for videos.
- Offer transcripts for audio content.
- Avoid auto-playing audio.
- Provide visual cues for important sound alerts (e.g. notifications).

Motor

Limited hand dexterity or inability to manipulate a mouse.



Common Challenges

- Difficulty using a mouse or precise pointer movements.
- May have difficulty clicking, dragging or typing.
- May rely on keyboard, switch devices or voice input.

Design Considerations

- Ensure all content is keyboard accessible.
- Provide clear focus indicators.
- Use large, easy-to-click buttons and links.
- Allow enough time to complete tasks (avoid time limits).

Cognitive

Learning disabilities and attention challenges requiring clear layout.



Common Challenges

- Difficulty understanding complex layout or language.
- May be easily distracted.
- Difficulty remembering information or completing multi-step tasks.

Design Considerations

- Use clear and consistent layout.
- Write in simple and plain language.
- Break information into small chunks with clear headings.
- Use consistent navigation and predictable design patterns.

Key Takeaway

Understanding different impairment categories helps us design web content that is **perceivable, operable, understandable** and **robust** for all users.

Color Contrast & Visibility

Good color contrast helps all users, including people with low vision and color blindness, to read and understand web content easily.

Use sufficient color contrast between text and background to ensure content is readable (WCAG).

1 What is Color Contrast?

Color contrast is the difference in brightness between text and background colors. Higher contrast makes text easier to read.

Good contrast improves readability for everyone.

Poor contrast makes text difficult to read.

2 Examples of Contrast

Accessible Text Contrast ratio: 12.6:1 Meets WCAG (≥ 4.5:1)	Low Contrast Contrast ratio: 1.8:1 Does not meet WCAG
Accessible Text Contrast ratio: 19.6:1 Meets WCAG (≥ 4.5:1)	Low Contrast Contrast ratio: 1.5:1 Does not meet WCAG
Accessible Text Contrast ratio: 21:1 Meets WCAG (≥ 4.5:1)	Low Contrast Contrast ratio: 3.0:1 Does not meet WCAG (for normal text)

3 WCAG Contrast Requirements

Text Type	Minimum Contrast Ratio
Normal text (less than 18 pt)	≥ 4.5 : 1
Large text (18 pt or bold 14 pt+)	≥ 3 : 1
UI components (icons, buttons, etc.)	≥ 3 : 1

Tip

- Use online contrast checkers to verify color combinations.
- Do not rely on color alone to convey information.
- Ensure text remains readable at different screen sizes and in various lighting conditions.

4 Consider Color Blindness

Some users have difficulty distinguishing certain colors. Choose color combinations that remain clear for everyone.

	Original	Simulated View
Normal Vision	Button	Button
Deuteranopia (green-blind)	Button	Button
Protanopia (red-blind)	Button	Button
Tritanopia (blue-blind)	Button	Button

Key Takeaway

Use sufficient color contrast and choose accessible color combinations to ensure your web content is **visible** and **readable** for all users.

Frequent Web Accessibility Flaws

Common issues that can make websites difficult or impossible for people with disabilities to use.

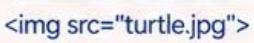
Avoiding these common flaws helps create web content that is more inclusive, usable and accessible for all.

1 Missing Alternative Text

Images without descriptive alt text cannot be understood by screen readers.

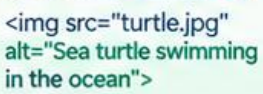
Problem





Improved





2 Poor Color Contrast

Text with low contrast is difficult to read, especially for users with low vision or color blindness.

Problem

This text is difficult to read because the contrast is too low.

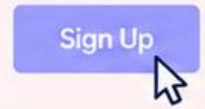
Improved

This text is easy to read with good color contrast.

3 No Keyboard Access

Links, buttons or menus that cannot be reached or operated using a keyboard.

Problem



Only works with a mouse.

Improved



Can be reached and activated using a keyboard (with clear focus indicator).

4 Missing Form Labels

Form fields without labels can be confusing for screen reader users.

Problem



No label
(screen readers do not know what to enter).

Improved



Clear and descriptive label for the form field.

5 Inaccessible Media

Videos or audio without captions or transcripts exclude users who are deaf or hard-of-hearing.

Problem



No captions provided.

Improved



Captions provided for all video content.

6 Poor Page Structure

Skipping headings or using headings only for visual styling makes content hard to navigate.

Problem



Headings are not used logically (just styled text).

Improved

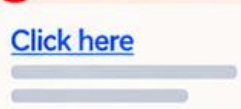


Logical heading structure helps all users navigate and understand the content.

7 Vague Link Text

Links with unclear text (e.g. “click here”) are not meaningful for screen reader users.

Problem



The purpose of the link is unclear.

Improved



Descriptive link text explains the destination.

8 Content That Relies Only on Color

Important information shown only by color can be missed by users with color blindness.

Problem



Error is indicated only by red color.

Improved



Error is indicated with both color and an icon (and text message).

Key Takeaway

Identify and fix these common accessibility flaws to make your website **more inclusive, usable and compliant** with web accessibility guidelines (e.g. WCAG).

Reliability & Information Evaluation

Evaluate online information critically to ensure it is accurate, trustworthy and relevant to your needs.

Not everything online is reliable. Use these strategies to assess the quality of information.

1 Source Credibility

Evaluate author credentials, publisher reputation, and domain backing.



What to check?

- Author's qualifications and expertise
- Publisher or organisation reputation
- Domain type (e.g. .edu, .gov, .org)
- Contact information and "About Us" page

2 Information Recency

Check publication and update dates to verify currency.



What to check?

- Publication date
- Last updated date
- Is the information still relevant?
- Be cautious with outdated content, especially for fast-changing topics (e.g. technology, health, policies).

3 Additional Resources

Cross-examine claims against primary research sources.

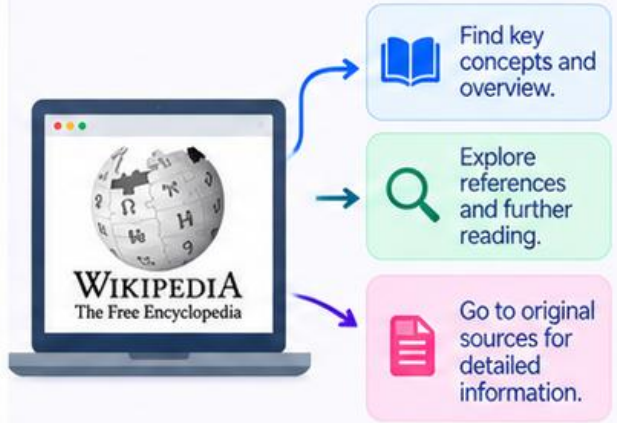


What to check?

- Look for primary sources (e.g. journal articles, official reports, statistics, government publications).
- Compare information from multiple reliable sources.
- Check if the evidence supports the claims made.
- Be wary of one-sided or unsupported claims.

4 Beyond Wikipedia

Use encyclopedias as launchpads, not final citations.



What to remember?

- Wikipedia is a useful starting point to understand a topic.
- Check the references section for reliable sources.
- Do not use Wikipedia as a final citation in academic work.
- Read the original sources for in-depth and credible information.

Key Takeaway

Be a **critical** and **informed** internet user. Evaluate the source, check the date, verify with reliable references, and use Wikipedia as a guide — **not a final source**.

QUICK TEST #1

Knowledge Check & Concept Review

QUIZ 1

Knowledge Review Questions

KNOWLEDGE CHECK

Question 1:

Describe the fundamental difference between the Internet and the Web.

KNOWLEDGE CHECK

Question 2:

Explain three major events that contributed to the commercialization and exponential growth of the Internet in the 1990s.

KNOWLEDGE CHECK

Question 3:

Is the concept of universal design important to web developers? Explain your answer.

The 7 Universal Design Principles

Designing products and environments to be usable by all people, to the greatest extent possible, regardless of age, ability or situation.

The 7 universal design principles help create **inclusive, accessible and usable** designs for everyone.

1 Equitable Use

The design is useful and accessible to people with diverse abilities.



Example
A website that works for everyone, including people with disabilities (e.g. screen reader support, keyboard navigation).

2 Flexibility in Use

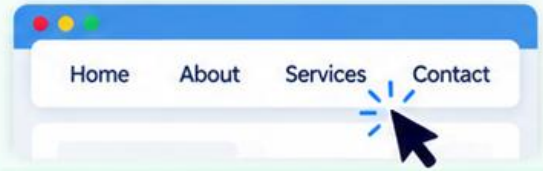
The design accommodates a wide range of individual preferences and abilities.



Example
A responsive website that can be used on a desktop, tablet or smartphone, and offers different input methods (e.g. touch, mouse, keyboard).

3 Simple and Intuitive Use

The design is easy to understand, regardless of the user's experience, knowledge, language skills or concentration level.



Example
Clear navigation and consistent icons that are easy to recognise (e.g. a house icon for Home).

4 Perceptible Information

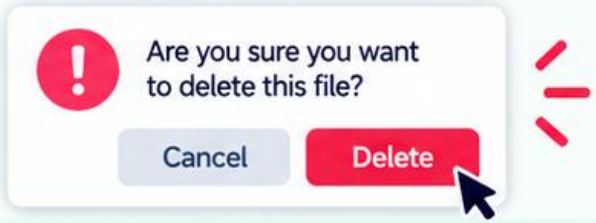
The design communicates necessary information effectively to all users, regardless of sensory abilities or environmental conditions.



Example
Use high contrast colours, clear text, alt text for images, captions for videos and readable fonts.

5 Tolerance for Error

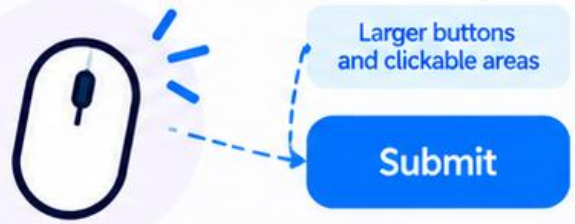
The design minimizes hazards and the adverse consequences of accidental or unintended actions.



Example
Show a confirmation message before deleting a file, and provide an option to undo action (e.g. "Undo").

6 Low Physical Effort

The design can be used efficiently and comfortably with a minimum of fatigue.



Example
Use large buttons and input fields that are easy to click or tap, reducing the need for precise or repetitive movements.

7 Size and Space for Approach and Use

Appropriate size and space are provided for approach, reach, manipulation and use, regardless of the user's body size, posture or mobility.



Example
Provide sufficient space between buttons and elements, and design layouts that can be easily used by people with different body sizes or mobility devices (e.g. wheelchair users).

ACTIVITY

Instructions: Accessibility Case Study

Step 1: Pair Up

Work in assigned student pairs to analyze a selected legacy website.

Step 2: Understand Principles

Review the 7 principles of Universal Design.

Step 3: Analyze Site

Spend 15-20 minutes inspecting for issues like poor color contrast, missing alt text, difficult navigation, small font sizes, or non-mobile-friendly design.

Step 4: Propose Solutions

Document violations and submit actionable UX recommendations.

CASE STUDIES

10 Websites for Accessibility Analysis

Here are 10 websites that would work well for your Universal Design Case Study activity:

1. **Ling's Cars** (www.lingscars.com)
2. **Arngren** (www.arngren.net)
3. **Yale School of Art** (art.yale.edu)
4. **Suzanne Collins Books** (www.suzannecollinsbooks.com)
5. **The World's Worst Website** (www.theworldsworstwebsiteever.com)
6. **Budgets Are Sexy** (www.budgetsaresexy.com)
7. **Rudgwick Steam Show** (www.rudgwicksteamshow.co.uk)
8. **Pacific Northwest X-Ray** (www.pnwx.com)
9. **Yvette's Bridal Formal** (www.yvettesbridalformal.com)
10. **Epicurrence** (www.epicurrence.com)

Computer Networks

A computer network is a collection of interconnected devices that can communicate and share data, resources and services with each other.

Computer networks enable people and devices to **connect, communicate, collaborate** and access information anywhere in the world.

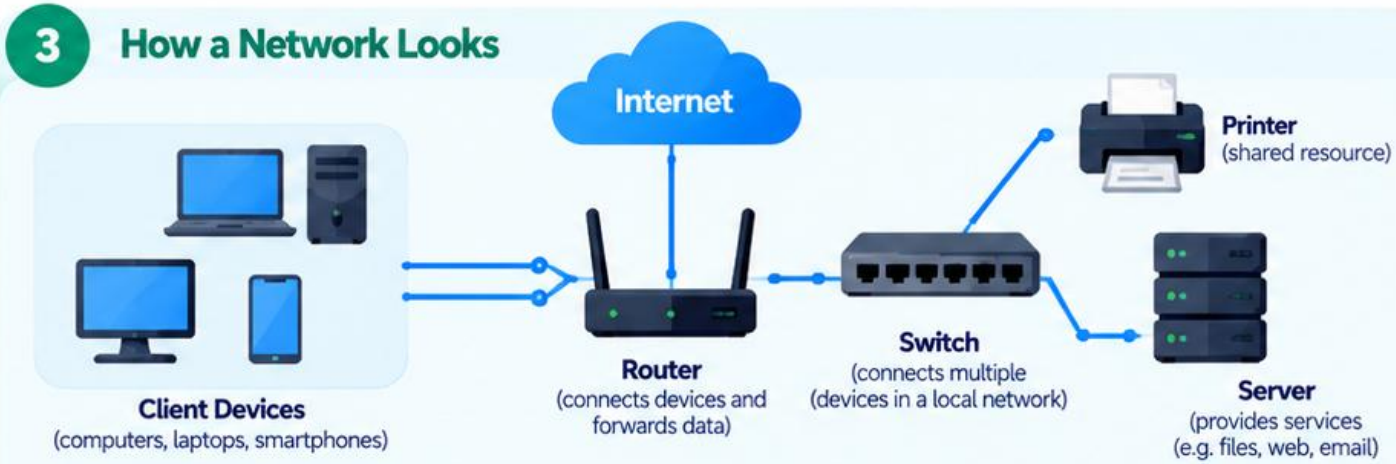
1 What is a Computer Network?

A computer network is a group of computers and other devices connected together using communication technologies so that they can **exchange data** and **share resources**.

2 Why are Networks Important?

- Share information
- Collaborate easily
- Share resources (e.g. printers, files)
- Access online services (e.g. web, email)
- Improve efficiency and productivity

3 How a Network Looks



4 Types of Computer Networks

PAN
(Personal Area Network)



Short-range network around a person (e.g. Bluetooth).

LAN
(Local Area Network)



Connects devices in a small area (e.g. home, office, school).

MAN
(Metropolitan Area Network)



Connects multiple LANs within a city or large campus.

WAN
(Wide Area Network)



Connects networks across large distances (e.g. countries, continents). The Internet is the largest WAN.

5 Key Components

End Devices



Send and receive data (e.g. PCs, smartphones).

Network Devices



Connect devices and forward data (e.g. router, switch, access point).

Transmission Media



Carry data between devices (e.g. cables, Wi-Fi, fiber optic).

Software & Protocols



Define the rules for communication (e.g. TCP/IP, HTTP).

6 How It Works

1. **Data is created**



A device creates data to be sent (e.g. a web request).

2. **Data is divided into packets**



Data is split into small packets for transmission.

3. **Packets are transmitted**



Packets travel through networks using routers and communication media.

4. **Data is received and reassembled**



Packets are reassembled at the destination device.

7 Common Network Protocols

TCP/IP	Provides reliable communication and addressing across networks.
HTTP / HTTPS	Used to access web pages.
FTP	Used to transfer files.
SMTP / IMAP / POP3	Used for sending and receiving emails.
DNS	Translates domain names to IP addresses.

The Client / Server Model

A computing model where clients request services or resources, and servers provide and manage those services.

The client/server model is widely used on the Internet, where many clients access services from a central server.

How It Works

Client

Requests a service or resource (e.g. a web page).



Client devices
(e.g. computer, tablet, smartphone)

1 Send a request
Client sends a request to the server (e.g. using HTTP).

Internet
Network that connects clients and servers.

2 Send a response
Server sends the requested resource back to the client (e.g. HTML, images, data).

Server

Processes the request and provides the requested resource (e.g. web page, data or file).



- Web pages
- Databases
- Files
- Applications

Server
(e.g. web server, database server)

Real-World Analogy



Client (Customer)
Places an order (requests a service).

Order (request)

Food (response)



Server (Chef)
Prepares the food (provides the service).

1 Client Initiates a Request

The client (e.g. web browser) sends a request to the server (e.g. for a web page).



Example:
A browser requests the homepage of a website.

2 Request Sent Over the Network

The request is sent over the Internet using a protocol such as HTTP.



Example:
The HTTP request is sent through networks and routers to reach the server.

3 Server Processes the Request

The server receives the request, processes it (e.g. runs application code, retrieves data from a database) and prepares the response.



Example:
A web server runs a script (e.g. PHP) and retrieves the requested web page.

4 Response Sent Back to Client

The server sends the response (e.g. HTML, CSS, images, data) back to the client over the Internet.



Example:
The browser receives and displays the web page.

Key Characteristics

Roles	Clients request services; servers provide services.
Uses protocols	Communicate using standard protocols (e.g. HTTP, FTP, SMTP).
Centralized resources	Servers store and manage data, files and applications.
Multiple clients	Many clients can connect to the same server simultaneously.
Scalable and reliable	Servers can be designed to handle many requests and provide high availability.

Web Client vs. Web Server

The web client requests a web page, and the web server processes the request and sends back the response.

The web client and web server work together using the **HTTP protocol** to deliver web pages on the Internet.

Web Client (Client-side)



- Runs on the user's device (e.g. computer, tablet, smartphone).
- Uses a web browser (e.g. Chrome, Firefox, Edge) to access web pages.
- Sends an HTTP request to the web server.

1 HTTP Request

Client requests a web page (e.g. https://www.umt.edu.my)

Internet

2 HTTP Response

Server sends the requested web page (e.g. HTML, CSS, images).

Web Server (Server-side)



- Runs on a server (physical or virtual) on the Internet.
- Processes the HTTP request (e.g. finds the requested file, runs application code).
- Sends the HTTP response containing the requested content (e.g. HTML, CSS, images).

Web Client in Detail

What it is	A software application (web browser) used by the user to request and view web pages.
Examples	Google Chrome, Mozilla Firefox, Microsoft Edge, Safari.
Main tasks	• Send HTTP requests (e.g. for a web page). • Display web content (HTML, CSS, images). • Handle user interaction (click, type, submit).
Devices	Desktop, laptop, tablet, smartphone, etc.

Web Server in Detail

What it is	A computer and software that stores web files and delivers them to clients over the Internet.
Examples	Apache HTTP Server, Nginx, Microsoft IIS, LiteSpeed.
Main tasks	• Receive and process HTTP requests. • Find and deliver the requested content. • Run server-side scripts (e.g. PHP, Python). • Manage security, performance and access.
Stored content	HTML files, CSS files, JavaScript files, images, videos, and server-side applications (e.g. PHP, Python, Java).

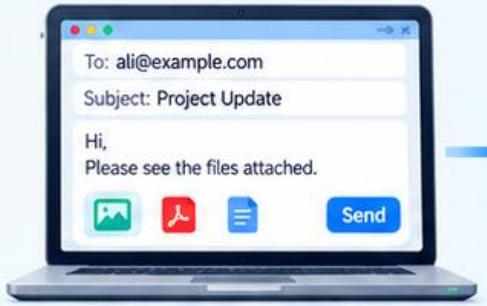
Key Differences

Aspect	Web Client	Web Server
Role	Requests and displays web content.	Stores, processes and delivers web content.
Location	User's device (client-side).	Server (usually in a data center or cloud).
Software examples	Chrome, Firefox, Edge, Safari.	Apache, Nginx, IIS, LiteSpeed.
Main action	Sends HTTP request.	Sends HTTP response.
Content handled	Displays HTML, CSS, images, etc.	Stores and serves HTML, CSS, images, etc. (and may run server-side code).
Type	Application software.	Server software + hardware.

MIME (Multi-Purpose Internet Mail Extensions)

A standard that extends email to support different types of content (e.g. text, images, audio, video, documents).

MIME allows emails to include content beyond **plain text** by describing the type and format of each part of the message.



Sender

Creates an email with text, images, attachments, etc.

SMTP

Sends the email (with MIME formatted content)

MIME Email Content

MIME Headers — Describe the type of content, format and structure of the email.

MIME Body (Multi-part)

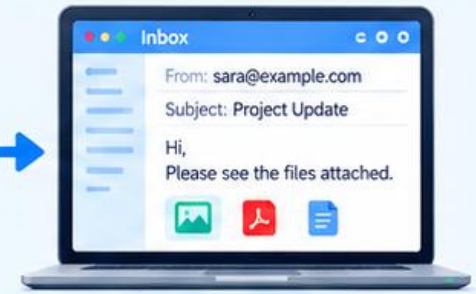
Can contain multiple parts with different content types.

Text	Image	Document	Video	...
text/plain	image/jpeg	application/pdf	video/mp4	



Mail Server

Stores and forwards the MIME email.



Receiver

Email client uses MIME information to display and handle each part correctly (text, images, attachments).

1 Key Points about MIME

Extends email format
Allows email to include non-text content (images, audio, video, documents).

Uses content types
Identifies the type and format of each part using a MIME type (e.g. text/plain, image/jpeg).

Supports multi-part messages
An email can contain multiple parts (e.g. plain text + HTML + attachments).

Widely used
Used by email clients and mail servers via SMTP, POP3 and IMAP.

2 Common MIME Types

Content Type	MIME Type	Description
	text/plain	Plain text
	text/html	HTML content
	image/jpeg	JPEG image
	image/png	PNG image
	application/pdf	PDF document
	application/msword	Microsoft Word document
	audio/mpeg	MP3 audio
	video/mp4	MP4 video

3 MIME Headers (Example)

```
MIME-Version: 1.0                          ← Indicates MIME is used.
Content-Type: multipart/mixed;
              boundary="abc123"            ← Specifies the content type and boundary.
From: sara@example.com
To: ali@example.com
Subject: Project Update

--abc123                                   ← Boundary between parts.
Content-Type: text/plain; charset=UTF-8
Content-Transfer-Encoding: 7bit

Hi,
Please see the files attached.

--abc123
Content-Type: image/jpeg
Content-Transfer-Encoding: base64
/9j/4AAQSkZJRgABAQAAAQABAAD....
(image data)

--abc123--                                 ← End of message.
```

4 MIME in Email Services

SMTP (Sending)
- Used to send MIME-formatted emails.
- Carries headers and all parts of the message.

POP3 (Receiving)
- Downloads the MIME email to the client.
- Client uses MIME information to display the content correctly.

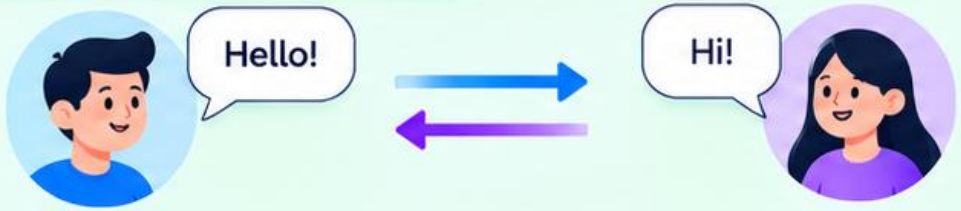
IMAP (Receiving)
- Keeps MIME-structured emails on the server.
- Client reads and displays MIME content (text, images, attachments, etc.).

What is a Protocol?

A protocol is a set of rules that define how data is communicated between devices over a network.

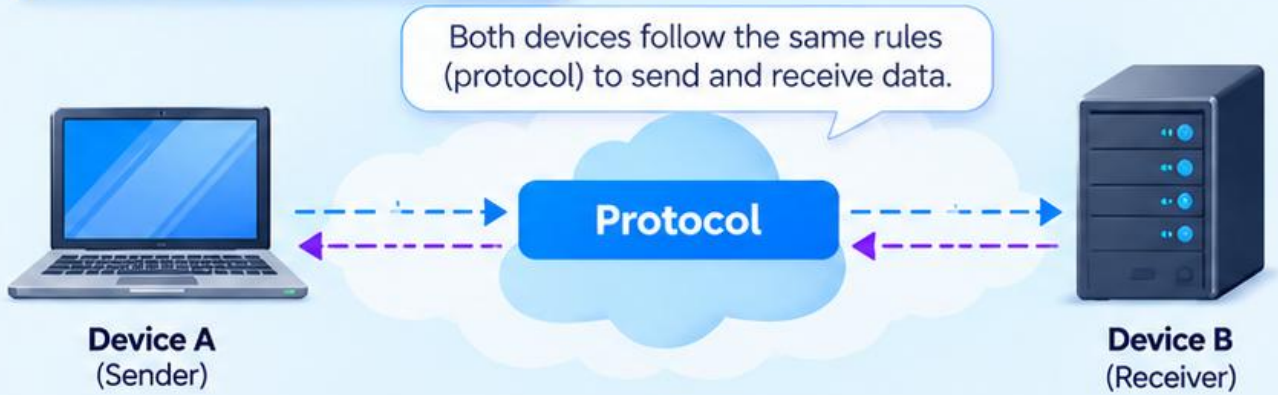
A protocol ensures that devices can **understand** each other, **exchange** data correctly, and **work together**.

Real-World Analogy



When people communicate, we follow common rules (e.g. language, grammar, turn-taking) so that both sides understand each other. A **protocol** works in the same way for devices.

In a Computer Network



1 Key Characteristics

Set of rules
Defines the format, order and meaning of messages.

Standardized
So that different devices and software can communicate.

Controls communication
How data is sent, received, and handled (e.g. error checking, sequencing).

Enables interoperability
Allows devices from different manufacturers to work together.

2 How a Protocol Works

1. **Format**
   Defines the structure of the data (e.g. headers, message format).
2. **Transmission**
   Determines how data is sent over the network (e.g. in packets).
3. **Error handling**
   Includes rules to detect and correct errors if needed.
4. **Completion**
   Ensures data is received correctly and in the right order.

3 Examples of Protocols

HTTP	Transfers web pages on the Internet (e.g. when you browse a website).
SMTP	Sends emails from your email client to a mail server.
POP3 / IMAP	Receives emails from a mail server to your email client.
FTP	Transfers files between a client and a server.
TCP/IP	Provides reliable delivery and addressing for data across networks.

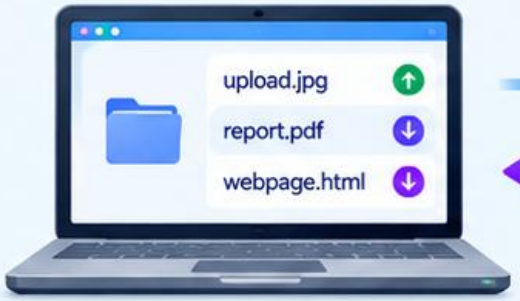
File Transfer Protocol (FTP)

A protocol for transferring files between a client and a server over a network.

FTP allows users to upload, download, manage and organize files on a remote server.

FTP Client

Your computer or application
(e.g. FileZilla, browser, command line)



Upload files to server
(STOR command)

Download files from server
(RETR command)

FTP Server

Stores files and manages access (e.g. web hosting server)



Remote Files

- /public_html
- /images
- /documents

What Can You Do with FTP?

- **Upload files**
  e.g. website files to a web server
- **Download files**
  e.g. backup or retrieve files
- **Manage files**
  create, delete, rename, move
- **Access remote directories**
  organize files on the server

1 How FTP Works

1. **Connect to the server**
   The client connects to the FTP server using an FTP client.
2. **Authenticate**
   Provide a username and password to access the server.
3. **Transfer files**
   Use commands to upload (STOR) or download (RETR) files.
4. **Manage files**
   List, delete, rename or move files on the server.
5. **Disconnect**
   Close the connection when finished.

2 FTP Connections and Ports

Control Connection

- Used for sending commands and receiving responses.
- Default port: **21**
- Stays open during the session.

Data Connection

- Used for transferring files and directory listings.
- Default port: **20** (Active mode)
- In Passive mode, the server provides a dynamic port.

3 FTP Modes

Active Mode



- Server initiates the data connection from port 20 to the client.
- May be blocked by firewalls.

Passive Mode



- Client initiates both control and data connections.
- More firewall-friendly (commonly used).

4 Advantages and Limitations

Advantages

- Simple and widely supported
- Good for transferring large files
- Useful for website publishing and server management

Limitations

- Not encrypted (username, password and files sent in plain text)
- Less secure than SFTP or FTPS
- Can be blocked by firewalls or network restrictions

5 Example FTP Commands

```
ftp example.com          # connect to server
user username            # enter username
pass password            # enter password
ls                       # list files
get file.txt             # download a file
put file.txt             # upload a file
bye                      # disconnect
```

Email Protocols: Sending vs. Receiving

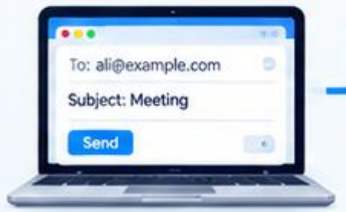
Different protocols are used to send and receive email.

Email uses different protocols: SMTP to send emails and POP3 or IMAP to receive emails.

Sending an Email

Uses SMTP to send the email from your email client to the recipient's mail server.

1 Compose Email
You write an email in an email client (e.g. Outlook, Gmail).



SMTP (send)

2 Outgoing Mail Server
Your email is sent to your mail server using SMTP.



Your Mail Server
(e.g. smtp.gmail.com)

3 Routes to Recipient
The email is routed across the Internet to the recipient's mail server.

Internet

4 Recipient's Mail Server
Your email is delivered to the recipient's mail server using SMTP.



Recipient's Mail Server
(e.g. mail.example.com)

5 Stored in Mailbox
The email is stored in the recipient's mailbox until it is retrieved.



Receiving an Email

Uses POP3 or IMAP to retrieve emails from the mail server to your email client.

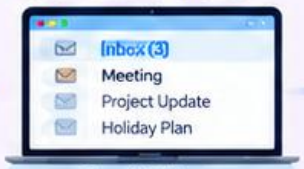
1 Email Stored on Server
New emails are stored in your mailbox on the mail server.



Your Mail Server
(e.g. mail.example.com)

POP3 or IMAP (receive)

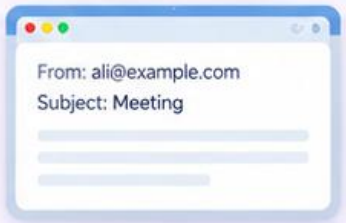
2 Retrieve Emails
Your email client uses POP3 or IMAP to connect to the mail server and retrieve emails.



Email Client
(e.g. Outlook, Gmail)

3 Download or Synchronize
POP3 downloads emails to your device, while IMAP keeps emails synchronized with the server.

4 Read Emails
You can read, reply or manage your emails in the email client.



SMTP (Simple Mail Transfer Protocol)

- Used for sending emails.
- Transfers emails from the sender's client to the recipient's mail server.
- Common port: 25 (server-to-server), 587 (submission), 465 (SSL/TLS).

POP3 (Post Office Protocol version 3)

- Used for receiving emails.
- Downloads emails from the server to your device (usually removes from server).
- Common ports: 110 (POP3), 995 (POP3 over SSL/TLS).

IMAP (Internet Message Access Protocol)

- Used for receiving emails.
- Keeps emails synchronized between your device and the server.
- Allows access from multiple devices.
- Common ports: 143 (IMAP), 993 (IMAP over SSL/TLS).

Key Differences

Feature	SMTP	POP3	IMAP
Purpose	Send emails	Receive emails	Receive emails
Stores on server	No	Usually no (removed)	Yes (synchronized)
Access from multiple devices	N/A	No (single device)	Yes
Typical ports	25, 587, 465	110, 995	143, 993

Hypertext Transfer Protocol (HTTP)

A protocol for transferring web pages and other resources on the World Wide Web.

HTTP defines how a client requests a resource and how a server responds.

Client (Web Browser)



Sends an HTTP request to a web server.

1 HTTP Request
e.g. Get a web page (index.html)

2 HTTP Response
e.g. Web page content (HTML, CSS, images)

Web Server



Processes the request and returns the resource.

Resources

Web pages · Images · Stylesheets · JavaScript files · Other files

1 How HTTP Works

1. The user enters a URL (e.g. https://www.umt.edu.my) in the browser.
2. The browser sends an **HTTP request** to the server for the resource.
3. The server processes the request.
4. The server sends an HTTP response containing the requested resource.
5. The browser displays the web page.

2 HTTP Request and Response

HTTP Request (from client)

Request Line

```
GET /index.html HTTP/1.1
Host: www.wmt.edu.my
User-Agent: Mozilla/5.0
Accept: text/html,application/xhtml+xml
```

HTTP Response (from server)

Status Code

```
HTTP/1.1 200 OK
Content-Type: text/html
Content-Length: 1256

<!DOCTYPE html>
<html> ... </html>
```

3 Common HTTP Methods

GET	Requests a resource (e.g. a web page).
POST	Sends data to the server (e.g. form submission).
PUT	Updates a resource (e.g. modify data).
DELETE	Deletes a resource.
HEAD	Requests only the headers (without the content).
OPTIONS	Returns the supported HTTP methods.

4 Common HTTP Status Codes

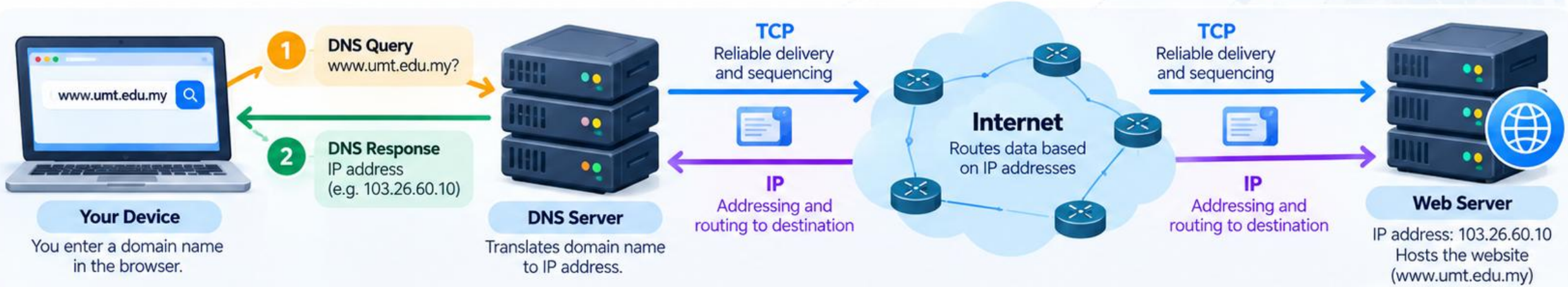
200	**OK** – Request successful.
301	**Moved Permanently** – The resource has a new URL.
404	**Not Found** – The requested resource was not found.
500	**Internal Server Error** – A server error occurred.
503	**Service Unavailable** – The server is temporarily unavailable.

5 HTTP Versions

HTTP/1.0	HTTP/1.1	HTTP/2
Introduced in 1996	Introduced in 1997	Introduced in 2015
Basic features	Persistent connections	Faster and more efficient
	Widely used	(e.g. multiplexing)

TCP / IP and Addressing

How data is sent, addressed and found on the Internet



1 TCP Role

Transmission Control Protocol

- Provides reliable, ordered delivery of data.
- Breaks data into smaller segments (packets).
- Checks for errors and requests retransmission if needed.
- Uses port numbers to identify applications (e.g. 80 for HTTP, 443 for HTTPS).



2 IP Role

Internet Protocol

- Provides logical addressing using IP addresses.
- Determines the best route for packets across networks.
- Delivers packets from the source IP address to the destination IP address.
- Works with TCP to send complete data to the correct device.



3 IPv4 Structure

32-bit address (4 octets)

192	168	1	10
Octet 1	Octet 2	Octet 3	Octet 4
Network part	Network part	Subnet part	Host part

- An IPv4 address is 32 bits long, written as four decimal numbers (0–255) separated by dots.
- Identifies a specific device on a network.
- Consists of a network part (which network) and a host part (which device).

Example:	192	168	1	10
	Network ID	Subnet ID		Host ID

4 DNS Link

Domain Name System

- Translates human-friendly domain names (e.g. www.umt.edu.my) to IP addresses (e.g. 103.26.60.10).
- Allows you to use easy-to-remember names instead of numeric IP addresses.
- Uses a hierarchical system with root servers, TLD servers (e.g. .my) and authoritative name servers.



Domain Names, URIs, and TLDs

From user-friendly names to complete web addresses

1 Domain Names

Human-friendly names used to identify websites on the Internet.

www.umt.edu.my

- **Subdomain** (optional) — www
- **Second-level domain (SLD)** — umt
- **Top-level domain (TLD)** — .edu.my

2 URIs (Uniform Resource Identifiers)

A URI is a complete address used to identify a resource on the Internet.

https://www.umt.edu.my/about

- **Scheme** — https (protocol)
- **Domain Name** — www.umt.edu.my (host)
- **Path** — /about (resource)

A domain name is part of a URI.
The URI also includes the scheme (e.g. https) and path (e.g. **/about**).

3 TLDs (Top-Level Domains)

The last part of a domain name that indicates the domain type or country.

www.umt.edu.my

TLDs help identify the purpose or geographic location of a website.

TLD — **.my**

Structure of a Domain Name

www . **umt** . **edu** . **my**

Subdomain (optional)	Second-level domain (SLD)	Domain category (indicates purpose)	Country code TLD (ccTLD)
Identifies a specific service or section. e.g. www, mail, portal	The unique name of the organization or entity. e.g. umt, google, moe	Common TLD used for educational institutions. e.g. edu, com, org	Indicates the country or region. e.g. my (Malaysia), uk (United Kingdom)

Common Types of TLDs

Generic TLDs (gTLDs)

Indicate the type or purpose of the organization.

.com	Commercial organizations
.org	Non-profit organizations
.edu	Educational institutions
.gov	Government organizations
.net	Network-related organizations

Country Code TLDs (ccTLDs)

Indicate a specific country or territory.

.my	Malaysia
.sg	Singapore
.uk	United Kingdom
.us	United States
.jp	Japan

Evolution of Markup Languages

From simple document markup to rich, structured and extensible languages for the modern web.

1960s

SGML

Standard Generalized Markup Language



A metalanguage for defining markup languages.

Key Feature

Flexible, extensible and platform independent.

1986

HTML

HyperText Markup Language

```
<html>
  <head>
  </head>
  <body>
  </body>
</html>
```

Designed for creating web pages on the early web.

Key Feature

Simple markup for structuring web content.

1998

XML

eXtensible Markup Language

```
<book>
  <title>...</title>
  <author>...</author>
</book>
```

A flexible markup language for structured data exchange.

Key Feature

Extensible and self-descriptive tags.

2004

XHTML

Extensible HTML

```
<html
  xmlns="http://www
.w3.org/1999/xhtml">
  <body>
  </body>
</html>
```

A stricter, XML-based version of HTML.

Key Feature

Combines HTML with XML rules for well-formed markup.

2008

HTML5

HyperText Markup Language 5

```
<!DOCTYPE html>
<html>
  <head>...</head>
  <body>
    <div>...</div>
  </body>
</html>
```

Introduced new elements and features for modern web applications.

Key Feature

Richer semantics, multimedia, graphics and better support for modern devices.

Today

Modern Markup Languages

e.g. HTML5 variants, SVG, MathML, etc.



Markup languages continue to evolve to support new web technologies and applications.

Key Feature

Specialized markup languages for graphics, mathematics and modern web ecosystems.

SUMMARY & QUICK TEST #2

Final Knowledge Review

KNOWLEDGE CHECK

Question 1:

Describe the components of the client/server model as applied to the Internet.

KNOWLEDGE CHECK

Question 2:

Identify two protocols used on the Internet to convey information that do not use the Web.

KNOWLEDGE CHECK

Question 3:

Explain the similarities and differences between a URL and a domain name.